

Wu Li

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EDUCATION

University of Massachusetts Lowell
Bachelor of Science in Computer Science

Lowell, MA
May 2025

WORK EXPERIENCE

Massachusetts Department of Transportation
GIS AI Engineer

Boston, MA
March 2026 - Current

- Trained and optimized AI image-classification models on georeferenced engineering drawings, achieving 90% detection confidence and generating GIS-ready spatial features.
- Developed workflows to convert engineering plan archives into machine-learning training datasets through georeferencing and image preprocessing.
- Designed and tested a GIS workflow to calculate roadway right-of-way widths, identify anomalous segments, and support quality assurance through automated spatial analysis.

Cambridge Strength
Web Developer

Arlington, MA
March 2025 - May 2025

- Developed and deployed a custom WordPress theme, streamlined member access and on-site communication for more than 240 concurrent members.
- Integrated PushPress backend to securely manage payments, memberships, and client data, enabling seamless billing and automated account syncing.

PROJECTS

Parking Lot Occupancy Detector - *GIS, ML, Ultralytics, ArcPy/ArcGIS Pro, OpenCV*

- A GIS project with an end-to-end pipeline that detects vehicles in aerial parking lot imagery and computes occupancy using computer vision and GIS analysis.

VisionScan AI - *AI/ML, AWS, JavaScript, React, TensorFlow.js, Node.js, Express.js, Git*

- Designed and deployed a full-stack AI web application to perform object detection through a webcam with more than 80 object categories using AWS, Vercel, and Render for scalable real-time monitoring.

Retnify - *Medical Imaging, ML, Python, PyTorch, ResNet-50, VGG16, OpenCV, NumPy*

- Built a CNN-based model to classify 84,484 OCT retinal images into 4 disease categories, achieving 97.2% validation accuracy and 0.972 weighted F1 score.

SKILLS & TECHNICAL TOOLS

Languages: JavaScript, TypeScript, Python, C, C++, Java, OCaml, SQL

Web Technologies: React, Next.js, Tailwind CSS, WordPress, ArcGIS JavaScript SDK

GIS & Data Technologies: PostGIS, PostgreSQL, ArcPy, ArcGIS Online, ArcGIS Pro

Developer Tools: VSCode, Git, Github, Postman, Docker, Node.js, Express, REST APIs